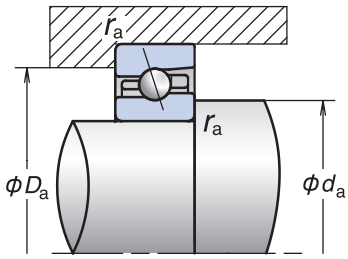
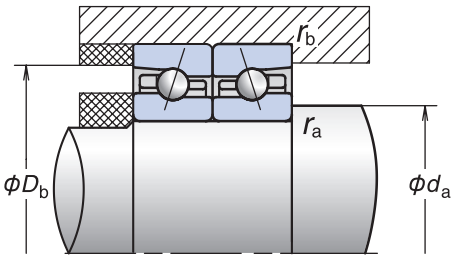


스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
86	104	105	1	0.5	104	195	503	986	63	81	125	173	− 4	− 9	− 21	− 34	0.377
86	104	105	1	0.5	111	215	573	1 136	72	94	145	202	− 4	− 9	− 21	− 34	0.337
86	104	105	1	0.5	182	366	792	1 603	160	207	278	370	− 4	− 8	− 15	− 25	0.381
86	104	105	1	0.5	201	415	913	1 867	186	242	326	436	− 4	− 8	− 15	− 25	0.341
86	104	105	1	0.5	49	190	478	–	53	86	123	–	0	− 8	− 19	–	0.366
86	104	105	1	0.5	49	210	545	–	60	99	143	–	0	− 8	− 19	–	0.326
86	104	105	1	0.5	49	210	545	–	60	99	143	–	0	− 8	− 19	–	0.326
86	104	105	1	0.5	49	309	646	–	89	166	217	–	0	− 8	− 15	–	0.366
86	104	105	1	0.5	49	349	743	–	100	194	254	–	0	− 8	− 15	–	0.326
86	104	105	1	0.5	49	349	743	–	100	194	254	–	0	− 8	− 15	–	0.326
86	104	105	1	0.5	49	190	478	–	53	86	123	–	0	− 8	− 19	–	0.425
86	104	105	1	0.5	49	210	545	–	60	99	143	–	0	− 8	− 19	–	0.385
86	104	105	1	0.5	49	210	545	–	60	99	143	–	0	− 8	− 19	–	0.385
86	104	105	1	0.5	49	309	646	–	89	166	217	–	0	− 8	− 15	–	0.425
86	104	105	1	0.5	49	349	743	–	100	194	254	–	0	− 8	− 15	–	0.385
86	104	105	1	0.5	49	349	743	–	100	194	254	–	0	− 8	− 15	–	0.385
87	118	120	1	0.6	202	382	921	1 880	78	102	151	215	− 6	− 14	− 31	− 52	0.855
87	118	120	1	0.6	215	422	1 045	2 162	89	118	176	251	− 6	− 14	− 31	− 52	0.736
87	118	120	1	0.6	345	624	1 513	2 903	198	246	345	448	− 6	− 11	− 23	− 37	0.880
87	118	120	1	0.6	381	704	1 744	3 377	229	286	404	528	− 6	− 11	− 23	− 37	0.761
87	118	120	1	0.6	98	752	1 762	3 060	170	340	464	573	0	− 10	− 20	− 30	0.864
87	118	120	1	0.6	98	327	611	–	72	111	141	–	0	− 10	− 19	–	0.921
87	118	120	1	0.6	98	359	688	–	80	127	164	–	0	− 10	− 19	–	0.867
87	118	120	1	0.6	98	359	688	–	80	127	164	–	0	− 10	− 19	–	0.867
87	118	120	1	0.6	98	623	1 272	–	118	225	292	–	0	− 12	− 22	–	0.921
87	118	120	1	0.6	98	704	1 461	–	133	262	343	–	0	− 12	− 22	–	0.867
87	118	120	1	0.6	98	704	1 461	–	133	262	343	–	0	− 12	− 22	–	0.867
87	118	120	1	0.6	98	327	611	–	72	111	141	–	0	− 10	− 19	–	1.12
87	118	120	1	0.6	98	359	688	–	80	127	164	–	0	− 10	− 19	–	1.06
87	118	120	1	0.6	98	359	688	–	80	127	164	–	0	− 10	− 19	–	1.06
87	118	120	1	0.6	98	623	1 272	–	118	225	292	–	0	− 12	− 22	–	1.12
87	118	120	1	0.6	98	704	1 461	–	133	262	343	–	0	− 12	− 22	–	1.06
87	118	120	1	0.6	98	704	1 461	–	133	262	343	–	0	− 12	− 22	–	1.06
90	130	134	2	1	305	595	1 367	2 752	83	110	159	224	− 12	− 24	− 47	− 76	1.43
90	130	134	2	1	333	667	1 566	3 185	96	127	186	262	− 12	− 24	− 47	− 76	1.18
90	130	134	2	1	463	925	2 161	4 345	201	259	357	473	− 9	− 17	− 33	− 54	1.42
90	130	134	2	1	517	1 056	2 507	5 078	234	303	420	558	− 9	− 17	− 33	− 54	1.17
90	130	134	2	1	98	1 095	2 696	4 035	156	356	494	576	0	− 15	− 30	− 40	1.45

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

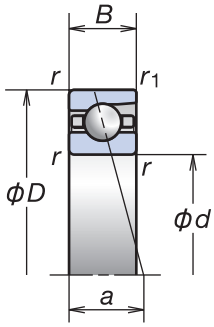
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

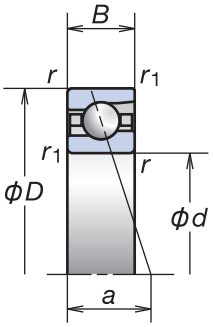
- 동등가 하중 …… P191
- 정등가 하중 …… P198
- 스페이서 · 노즐 목표 위치 …… P237
- 그리스 추천 봉입량 …… P257

1. 앵귤러 볼 베어링

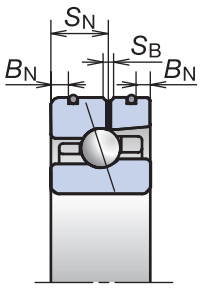
내경85mm



79,70,72계열
BNR10,20계열
BER10,20계열



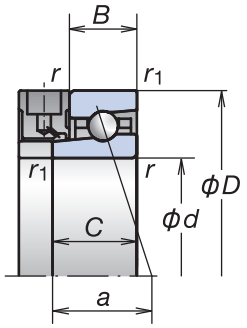
BNR19,29계열
BER19,29계열



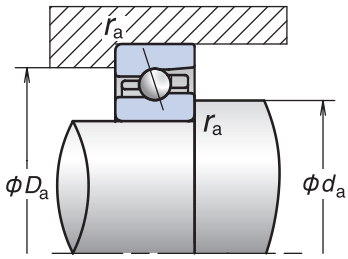
로바스트 쇼트

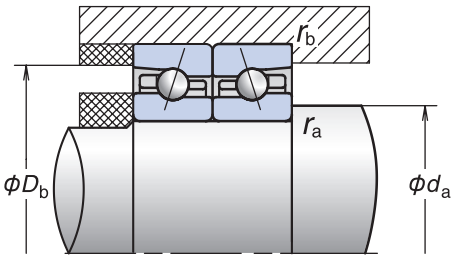
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7917C	85	120	18	–	–	–	–	1.1	0.6	41.0	40.5	25.9	15	16.5	22.7	11 300	17 100
*7917CSN24	85	120	18	–	–	–	–	1.1	0.6	(41.0)	(40.5)	30.7	15	16.5	22.7	14 700	22 300
*7917A5	85	120	18	–	–	–	–	1.1	0.6	38.5	38.5	30.0	25	–	32.9	9 800	14 700
*7917A5SN24	85	120	18	–	–	–	–	1.1	0.6	(38.5)	(38.5)	35.6	25	–	32.9	12 700	19 100
*85BNR19E	85	120	18	–	–	–	–	1.1	0.6	29.4	26.3	38.0	18	10.8	25.7	14 700	20 900
*85BNR19H	85	120	18	–	4.0	10.4	2.2	1.1	0.6	(29.4)	(26.3)	24.8	18	10.8	25.7	17 600	27 400
*85BNR19X	85	120	18	23	4.0	10.4	2.2	1.1	0.6	(29.4)	(26.3)	24.8	18	10.8	25.7	20 500	32 200
*85BER19E	85	120	18	–	–	–	–	1.1	0.6	28.1	25.2	35.5	25	–	32.9	12 600	17 800
*85BER19H	85	120	18	–	4.0	10.4	2.2	1.1	0.6	(28.1)	(25.2)	30.0	25	–	32.9	15 700	24 400
*85BER19X	85	120	18	23	4.0	10.4	2.2	1.1	0.6	(28.1)	(25.2)	30.0	25	–	32.9	18 600	29 300
*85BNR29EV1V	85	120	22	–	–	–	–	1.1	0.6	29.4	26.3	38.0	18	10.8	27.7	14 700	–
*85BNR29HV1V	85	120	22	–	–	–	–	1.1	0.6	(29.4)	(26.3)	24.8	18	10.8	27.7	17 600	–
*85BNR29XV1V	85	120	22	–	–	–	–	1.1	0.6	(29.4)	(26.3)	24.8	18	10.8	27.7	20 500	–
*85BER29EV1V	85	120	22	–	–	–	–	1.1	0.6	28.1	25.2	35.5	25	–	34.9	12 600	–
*85BER29HV1V	85	120	22	–	–	–	–	1.1	0.6	(28.1)	(25.2)	30.0	25	–	34.9	15 700	–
*85BER29XV1V	85	120	22	–	–	–	–	1.1	0.6	(28.1)	(25.2)	30.0	25	–	34.9	18 600	–
*7017C	85	130	22	–	–	–	–	1.1	0.6	63.0	58.5	38.0	15	15.9	25.4	10 700	16 300
*7017CSN24	85	130	22	–	–	–	–	1.1	0.6	(63.0)	(58.5)	45.0	15	15.9	25.4	14 000	21 300
*7017A5	85	130	22	–	–	–	–	1.1	0.6	60.0	55.5	43.0	25	–	36.1	9 400	14 000
*7017A5SN24	85	130	22	–	–	–	–	1.1	0.6	(60.0)	(55.5)	50.9	25	–	36.1	12 100	18 200
*7017A	85	130	22	–	–	–	–	1.1	0.6	57.5	53.5	33.0	30	–	42.0	7 000	9 400
*85BNR10E	85	130	22	–	–	–	–	1.1	0.6	26.8	25.7	37.5	18	11.0	28.5	14 000	20 000
*85BNR10H	85	130	22	–	4.7	12.2	2.2	1.1	0.6	(26.8)	(25.7)	24.5	18	11.0	28.5	16 800	26 100
*85BNR10X	85	130	22	27	4.7	12.2	2.2	1.1	0.6	(26.8)	(25.7)	24.5	18	11.0	28.5	19 600	30 700
*85BER10E	85	130	22	–	–	–	–	1.1	0.6	25.6	24.6	43.5	25	–	36.1	12 000	17 000
*85BER10H	85	130	22	–	4.7	12.2	2.2	1.1	0.6	(25.6)	(24.6)	29.5	25	–	36.1	14 900	23 300
*85BER10X	85	130	22	27	4.7	12.2	2.2	1.1	0.6	(25.6)	(24.6)	29.5	25	–	36.1	17 700	28 000
*85BNR20EV1V	85	130	27	–	–	–	–	1.1	0.6	26.8	25.7	37.5	18	11.0	31.0	14 000	–
*85BNR20HV1V	85	130	27	–	–	–	–	1.1	0.6	(26.8)	(25.7)	24.5	18	11.0	31.0	16 800	–
*85BNR20XV1V	85	130	27	–	–	–	–	1.1	0.6	(26.8)	(25.7)	24.5	18	11.0	31.0	19 600	–
*85BER20EV1V	85	130	27	–	–	–	–	1.1	0.6	25.6	24.6	43.5	25	–	38.6	12 000	–
*85BER20HV1V	85	130	27	–	–	–	–	1.1	0.6	(25.6)	(24.6)	29.5	25	–	38.6	14 900	–
*85BER20XV1V	85	130	27	–	–	–	–	1.1	0.6	(25.6)	(24.6)	29.5	25	–	38.6	17 700	–
7217C	85	150	28	–	–	–	–	2	1	113	90.5	60.5	15	14.7	29.7	9 800	14 900
7217CSN24	85	150	28	–	–	–	–	2	1	(113)	(90.5)	71.9	15	14.7	29.7	12 800	19 500
7217A5	85	150	28	–	–	–	–	2	1	107	86.5	70.0	25	–	41.4	8 600	12 800
7217A5SN24	85	150	28	–	–	–	–	2	1	(107)	(86.5)	83.1	25	–	41.4	11 100	16 600
7217A	85	150	28	–	–	–	–	2	1	104	83.5	53.5	30	–	47.9	6 400	8 600

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트 Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
92	113	115	1	0.6	138	307	629	1 281	71	98	135	191	− 6	− 14	− 25	− 41	0.534
92	113	115	1	0.6	150	345	719	1 481	81	114	158	224	− 6	− 14	− 25	− 41	0.470
92	113	115	1	0.6	227	427	950	1 909	176	222	300	396	− 5	− 9	− 17	− 28	0.541
92	113	115	1	0.6	253	486	1 099	2 228	205	259	352	467	− 5	− 9	− 17	− 28	0.477
92	113	115	1	0.6	49	193	652	–	55	88	140	–	0	− 8	− 24	–	0.527
92	113	115	1	0.6	49	214	749	–	61	102	164	–	0	− 8	− 24	–	0.456
92	113	115	1	0.6	49	214	749	–	61	102	164	–	0	− 8	− 24	–	0.456
92	113	115	1	0.6	49	317	898	–	91	171	248	–	0	− 8	− 19	–	0.527
92	113	115	1	0.6	49	358	1 039	–	102	199	292	–	0	− 8	− 19	–	0.456
92	113	115	1	0.6	49	358	1 039	–	102	199	292	–	0	− 8	− 19	–	0.456
92	113	115	1	0.6	49	193	652	–	55	88	140	–	0	− 8	− 24	–	0.617
92	113	115	1	0.6	49	214	749	–	61	102	164	–	0	− 8	− 24	–	0.554
92	113	115	1	0.6	49	214	749	–	61	102	164	–	0	− 8	− 24	–	0.554
92	113	115	1	0.6	49	317	898	–	91	171	248	–	0	− 8	− 19	–	0.617
92	113	115	1	0.6	49	358	1 039	–	102	199	292	–	0	− 8	− 19	–	0.554
92	113	115	1	0.6	49	358	1 039	–	102	199	292	–	0	− 8	− 19	–	0.554
92	123	125	1	0.6	205	393	995	1 956	81	106	161	224	− 6	− 14	− 32	− 52	0.898
92	123	125	1	0.6	219	434	1 130	2 252	93	122	187	262	− 6	− 14	− 32	− 52	0.773
92	123	125	1	0.6	305	646	1 487	2 915	196	257	353	462	− 5	− 11	− 22	− 36	0.904
92	123	125	1	0.6	334	729	1 713	3 390	226	299	413	544	− 5	− 11	− 22	− 36	0.779
92	123	125	1	0.6	98	780	1 837	3 196	176	355	486	600	0	− 10	− 20	− 30	0.907
92	123	125	1	0.6	98	334	627	–	73	114	146	–	0	− 10	− 19	–	0.962
92	123	125	1	0.6	98	367	707	–	82	132	170	–	0	− 10	− 19	–	0.906
92	123	125	1	0.6	98	367	707	–	82	132	170	–	0	− 10	− 19	–	0.906
92	123	125	1	0.6	98	640	1 311	–	122	232	303	–	0	− 12	− 22	–	0.962
92	123	125	1	0.6	98	723	1 508	–	136	271	355	–	0	− 12	− 22	–	0.906
92	123	125	1	0.6	98	723	1 508	–	136	271	355	–	0	− 12	− 22	–	0.906
92	123	125	1	0.6	98	334	627	–	73	114	146	–	0	− 10	− 19	–	1.16
92	123	125	1	0.6	98	367	707	–	82	132	170	–	0	− 10	− 19	–	1.11
92	123	125	1	0.6	98	367	707	–	82	132	170	–	0	− 10	− 19	–	1.11
92	123	125	1	0.6	98	640	1 311	–	122	232	303	–	0	− 12	− 22	–	1.16
92	123	125	1	0.6	98	723	1 508	–	136	271	355	–	0	− 12	− 22	–	1.11
92	123	125	1	0.6	98	723	1 508	–	136	271	355	–	0	− 12	− 22	–	1.11
95	140	144	2	1	355	697	1 658	3 358	90	119	176	250	− 14	− 27	− 53	− 85	1.79
95	140	144	2	1	391	786	1 907	3 898	104	139	206	290	− 14	− 27	− 53	− 85	1.47
95	140	144	2	1	530	1 095	2 431	4 882	217	283	383	507	− 10	− 19	− 35	− 57	1.79
95	140	144	2	1	595	1 255	2 823	5 711	253	331	450	598	− 10	− 19	− 35	− 57	1.47
95	140	144	2	1	98	1 135	3 473	5 794	161	370	558	679	0	− 15	− 35	− 50	1.80

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

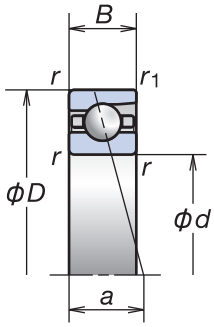
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

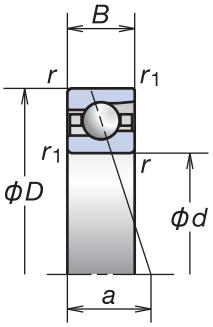
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

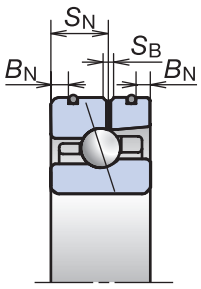
내경90mm



79,70,72계열
BNR10,20계열
BER10,20계열



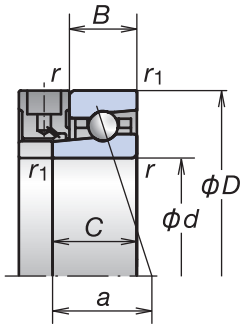
BNR19,29계열
BER19,29계열



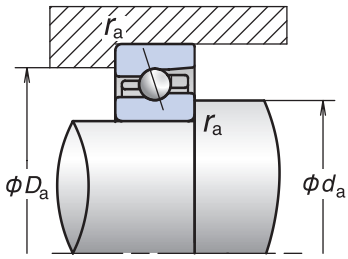
로바스트 쇼트

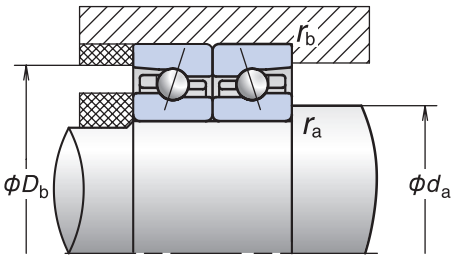
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7918C	90	125	18	–	–	–	–	1.1	0.6	44.0	46.0	29.1	15	16.6	23.4	10 700	16 300
*7918CSN24	90	125	18	–	–	–	–	1.1	0.6	(44.0)	(46.0)	34.6	15	16.6	23.4	14 000	21 300
*7918A5	90	125	18	–	–	–	–	1.1	0.6	41.0	43.5	33.5	25	–	34.1	9 400	14 000
*7918A5SN24	90	125	18	–	–	–	–	1.1	0.6	(41.0)	(43.5)	40.0	25	–	34.1	12 100	18 200
*90BNR19E	90	125	18	–	–	–	–	1.1	0.6	31.5	29.7	43.0	18	10.9	26.5	14 000	20 000
*90BNR19H	90	125	18	–	4.0	10.4	2.2	1.1	0.6	(31.5)	(29.7)	28.1	18	10.9	26.5	16 800	26 100
*90BNR19X	90	125	18	23	4.0	10.4	2.2	1.1	0.6	(31.5)	(29.7)	28.1	18	10.9	26.5	19 600	30 700
*90BER19E	90	125	18	–	–	–	–	1.1	0.6	30.0	28.5	50.5	25	–	34.1	12 000	17 000
*90BER19H	90	125	18	–	4.0	10.4	2.2	1.1	0.6	(30.0)	(28.5)	34.0	25	–	34.1	14 900	23 300
*90BER19X	90	125	18	23	4.0	10.4	2.2	1.1	0.6	(30.0)	(28.5)	34.0	25	–	34.1	17 700	28 000
*90BNR29EV1V	90	125	22	–	–	–	–	1.1	0.6	31.5	29.7	43.0	18	10.9	28.5	14 000	–
*90BNR29HV1V	90	125	22	–	–	–	–	1.1	0.6	(31.5)	(29.7)	28.1	18	10.9	28.5	16 800	–
*90BNR29XV1V	90	125	22	–	–	–	–	1.1	0.6	(31.5)	(29.7)	28.1	18	10.9	28.5	19 600	–
*90BER29EV1V	90	125	22	–	–	–	–	1.1	0.6	30.0	28.5	50.5	25	–	36.1	12 000	–
*90BER29HV1V	90	125	22	–	–	–	–	1.1	0.6	(30.0)	(28.5)	34.0	25	–	36.1	14 900	–
*90BER29XV1V	90	125	22	–	–	–	–	1.1	0.6	(30.0)	(28.5)	34.0	25	–	36.1	17 700	–
*7018C	90	140	24	–	–	–	–	1.5	1	75.5	69.0	44.5	15	15.7	27.4	10 000	15 300
*7018CSN24	90	140	24	–	–	–	–	1.5	1	(75.5)	(69.0)	53.2	15	15.7	27.4	13 100	19 900
*7018A5	90	140	24	–	–	–	–	1.5	1	71.0	65.5	52.0	25	–	38.8	8 700	13 100
*7018A5SN24	90	140	24	–	–	–	–	1.5	1	(71.0)	(65.5)	62.1	25	–	38.8	11 400	17 000
*7018A	90	140	24	–	–	–	–	1.5	1	68.5	63.5	40.5	30	–	45.2	6 600	8 700
*90BNR10E	90	140	24	–	–	–	–	1.5	1	35.0	33.0	48.0	18	10.9	30.7	13 100	18 700
*90BNR10H	90	140	24	–	5.5	14.5	2.2	1.5	1	(35.0)	(33.0)	31.5	18	10.9	30.7	15 700	24 400
*90BNR10X	90	140	24	29	5.5	14.5	2.2	1.5	1	(35.0)	(33.0)	31.5	18	10.9	30.7	18 300	28 700
*90BER10E	90	140	24	–	–	–	–	1.5	1	33.5	31.5	56.0	25	–	38.8	11 300	15 900
*90BER10H	90	140	24	–	5.5	14.5	2.2	1.5	1	(33.5)	(31.5)	38.0	25	–	38.8	14 000	21 800
*90BER10X	90	140	24	29	5.5	14.5	2.2	1.5	1	(33.5)	(31.5)	38.0	25	–	38.8	16 600	26 100
*90BNR20EV1V	90	140	30	–	–	–	–	1.5	1	35.0	33.0	48.0	18	10.9	33.7	13 100	–
*90BNR20HV1V	90	140	30	–	–	–	–	1.5	1	(35.0)	(33.0)	31.5	18	10.9	33.7	15 700	–
*90BNR20XV1V	90	140	30	–	–	–	–	1.5	1	(35.0)	(33.0)	31.5	18	10.9	33.7	18 300	–
*90BER20EV1V	90	140	30	–	–	–	–	1.5	1	33.5	31.5	56.0	25	–	41.8	11 300	–
*90BER20HV1V	90	140	30	–	–	–	–	1.5	1	(33.5)	(31.5)	38.0	25	–	41.8	14 000	–
*90BER20XV1V	90	140	30	–	–	–	–	1.5	1	(33.5)	(31.5)	38.0	25	–	41.8	16 600	–
7218C	90	160	30	–	–	–	–	2	1	129	105	72.0	15	14.6	31.7	9 200	14 000
7218CSN24	90	160	30	–	–	–	–	2	1	(129)	(105)	85.5	15	14.6	31.7	12 000	18 300
7218A5	90	160	30	–	–	–	–	2	1	123	100	83.5	25	–	44.1	8 000	12 000
7218A5SN24	90	160	30	–	–	–	–	2	1	(123)	(100)	99.2	25	–	44.1	10 400	15 600
7218A	90	160	30	–	–	–	–	2	1	118	96.5	64.5	30	–	51.1	6 000	8 000

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트 Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
97	118	120	1	0.6	153	289	740	1 488	79	102	156	219	− 3	− 9	− 23	− 39	0.568
97	118	120	1	0.6	159	314	834	1 703	89	117	180	255	− 3	− 9	− 23	− 39	0.496
97	118	120	1	0.6	272	500	1 096	2 184	203	253	341	449	− 4	− 8	− 16	− 27	0.560
97	118	120	1	0.6	296	560	1 255	2 531	233	294	399	528	− 4	− 8	− 16	− 27	0.488
97	118	120	1	0.6	98	282	711	–	75	109	156	–	0	− 8	− 21	–	0.552
97	118	120	1	0.6	98	308	804	–	84	125	181	–	0	− 8	− 21	–	0.480
97	118	120	1	0.6	98	308	804	–	84	125	181	–	0	− 8	− 21	–	0.480
97	118	120	1	0.6	98	432	977	–	124	206	276	–	0	− 8	− 17	–	0.552
97	118	120	1	0.6	98	482	1 117	–	139	239	323	–	0	− 8	− 17	–	0.480
97	118	120	1	0.6	98	482	1 117	–	139	239	323	–	0	− 8	− 17	–	0.480
97	118	120	1	0.6	98	282	711	–	75	109	156	–	0	− 8	− 21	–	0.653
97	118	120	1	0.6	98	308	804	–	84	125	181	–	0	− 8	− 21	–	0.582
97	118	120	1	0.6	98	308	804	–	84	125	181	–	0	− 8	− 21	–	0.582
97	118	120	1	0.6	98	432	977	–	124	206	276	–	0	− 8	− 17	–	0.653
97	118	120	1	0.6	98	482	1 117	–	139	239	323	–	0	− 8	− 17	–	0.582
97	118	120	1	0.6	98	482	1 117	–	139	239	323	–	0	− 8	− 17	–	0.582
99	131	134	1.5	0.8	247	502	1 187	2 373	87	117	172	241	− 8	− 18	− 37	− 60	1.16
99	131	134	1.5	0.8	266	560	1 355	2 741	99	135	200	282	− 8	− 18	− 37	− 60	0.994
99	131	134	1.5	0.8	409	779	1 758	3 498	218	275	374	494	− 7	− 13	− 25	− 41	1.17
99	131	134	1.5	0.8	454	886	2 031	4 079	252	321	440	582	− 7	− 13	− 25	− 41	1.00
99	131	134	1.5	0.8	98	782	2 483	3 977	176	356	543	650	0	− 10	− 25	− 35	1.18
99	131	134	1.5	0.8	98	338	830	–	75	116	164	–	0	− 10	− 24	–	1.24
99	131	134	1.5	0.8	98	372	943	–	83	134	191	–	0	− 10	− 24	–	1.16
99	131	134	1.5	0.8	98	372	943	–	83	134	191	–	0	− 10	− 24	–	1.16
99	131	134	1.5	0.8	98	653	1 339	–	124	238	309	–	0	− 12	− 22	–	1.24
99	131	134	1.5	0.8	98	739	1 541	–	139	277	362	–	0	− 12	− 22	–	1.16
99	131	134	1.5	0.8	98	739	1 541	–	139	277	362	–	0	− 12	− 22	–	1.16
99	131	134	1.5	0.8	98	338	830	–	75	116	164	–	0	− 10	− 24	–	1.52
99	131	134	1.5	0.8	98	372	943	–	83	134	191	–	0	− 10	− 24	–	1.44
99	131	134	1.5	0.8	98	372	943	–	83	134	191	–	0	− 10	− 24	–	1.44
99	131	134	1.5	0.8	98	653	1 339	–	124	238	309	–	0	− 12	− 22	–	1.52
99	131	134	1.5	0.8	98	739	1 541	–	139	277	362	–	0	− 12	− 22	–	1.44
99	131	134	1.5	0.8	98	739	1 541	–	139	277	362	–	0	− 12	− 22	–	1.44
100	150	154	2	1	384	771	1 865	3 713	95	126	187	262	− 15	− 29	− 57	− 90	2.20
100	150	154	2	1	425	872	2 150	4 316	109	146	219	308	− 15	− 29	− 57	− 90	1.80
100	150	154	2	1	658	1 272	2 899	5 945	240	304	416	556	− 12	− 21	− 39	− 64	2.31
100	150	154	2	1	744	1 462	3 375	6 965	279	357	490	657	− 12	− 21	− 39	− 64	1.91
100	150	154	2	1	98	1 676	4 314	5 954	165	434	616	697	0	− 20	− 40	− 50	2.23

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

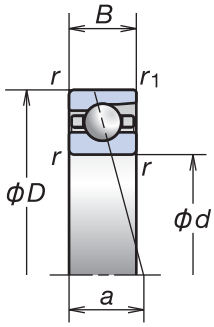
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

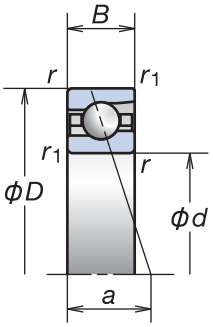
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

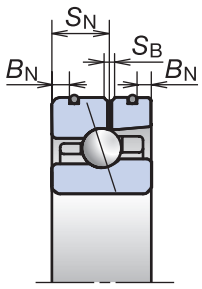
내경95mm



79,70,72계열
BNR10,20계열
BER10,20계열



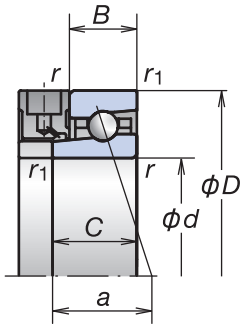
BNR19,29계열
BER19,29계열



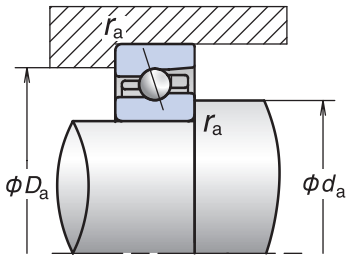
로바스트 쇼트

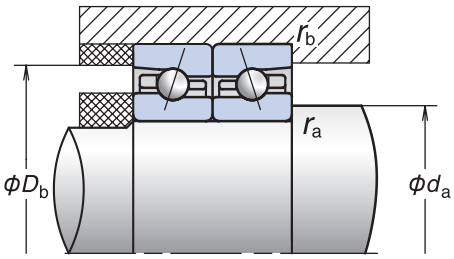
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7919C	95	130	18	–	–	–	–	1.1	0.6	44.5	48.0	30.0	15	16.7	24.1	10 300	15 600
*7919CSN24	95	130	18	–	–	–	–	1.1	0.6	(44.5)	(48.0)	35.8	15	16.7	24.1	13 400	20 300
*7919A5	95	130	18	–	–	–	–	1.1	0.6	42.0	45.5	35.0	25	–	35.2	8 900	13 400
*7919A5SN24	95	130	18	–	–	–	–	1.1	0.6	(42.0)	(45.5)	41.5	25	–	35.2	11 600	17 400
95BNR19E	95	130	18	–	–	–	–	1.1	0.6	32.0	31.0	50.0	18	10.9	27.3	13 400	19 100
95BNR19H	95	130	18	–	4.0	10.4	2.2	1.1	0.6	(32.0)	(31.0)	32.5	18	10.9	27.3	16 000	24 900
95BNR19X	95	130	18	23	4.0	10.4	2.2	1.1	0.6	(32.0)	(31.0)	32.5	18	10.9	27.3	18 700	29 400
95BER19E	95	130	18	–	–	–	–	1.1	0.6	30.5	29.7	58.5	25	–	35.2	11 500	16 200
95BER19H	95	130	18	–	4.0	10.4	2.2	1.1	0.6	(30.5)	(29.7)	39.5	25	–	35.2	14 300	22 300
95BER19X	95	130	18	23	4.0	10.4	2.2	1.1	0.6	(30.5)	(29.7)	39.5	25	–	35.2	16 900	26 700
7019C	95	145	24	–	–	–	–	1.5	1	77.0	73.0	47.0	15	15.9	28.1	9 600	14 600
7019CSN24	95	145	24	–	–	–	–	1.5	1	(77.0)	(73.0)	55.8	15	15.9	28.1	12 500	19 000
7019A5	95	145	24	–	–	–	–	1.5	1	73.0	69.5	52.5	25	–	40.0	8 400	12 500
7019A5SN24	95	145	24	–	–	–	–	1.5	1	(73.0)	(69.5)	62.7	25	–	40.0	10 900	16 300
7019A	95	145	24	–	–	–	–	1.5	1	70.0	67.0	40.5	30	–	46.6	6 300	8 400
95BNR10E	95	145	24	–	–	–	–	1.5	1	35.5	34.5	50.0	18	10.8	31.3	12 500	17 900
95BNR10H	95	145	24	–	5.5	14.5	2.2	1.5	1	(35.5)	(34.5)	32.5	18	10.8	31.3	15 000	23 400
95BNR10X	95	145	24	29	5.5	14.5	2.2	1.5	1	(35.5)	(34.5)	32.5	18	10.8	31.3	17 500	27 500
95BER10E	95	145	24	–	–	–	–	1.5	1	34.0	33.0	58.5	25	–	39.7	10 800	15 200
95BER10H	95	145	24	–	5.5	14.5	2.2	1.5	1	(34.0)	(33.0)	39.5	25	–	39.7	13 400	20 900
95BER10X	95	145	24	29	5.5	14.5	2.2	1.5	1	(34.0)	(33.0)	39.5	25	–	39.7	15 900	25 000
7219C	95	170	32	–	–	–	–	2.1	1.1	139	112	76.0	15	14.6	33.7	8 700	13 300
7219CSN24	95	170	32	–	–	–	–	2.1	1.1	(139)	(112)	90.0	15	14.6	33.7	11 400	17 300
7219A5	95	170	32	–	–	–	–	2.1	1.1	133	107	87.0	25	–	46.9	7 600	11 400
7219A5SN24	95	170	32	–	–	–	–	2.1	1.1	(133)	(107)	103	25	–	46.9	9 900	14 800
7219A	95	170	32	–	–	–	–	2.1	1.1	128	103	67.0	30	–	54.2	5 700	7 600

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
102	123	125	1	0.6	154	294	800	1 588	81	105	164	230	− 3	− 9	− 24	− 40	0.597
102	123	125	1	0.6	161	320	903	1 819	91	121	191	268	− 3	− 9	− 24	− 40	0.522
102	123	125	1	0.6	226	512	1 218	2 371	195	261	363	475	− 3	− 8	− 17	− 28	0.603
102	123	125	1	0.6	244	574	1 397	2 751	223	304	425	558	− 3	− 8	− 17	− 28	0.528
102	123	125	1	0.6	98	288	775	–	77	112	164	–	0	− 8	− 22	–	0.571
102	123	125	1	0.6	98	314	878	–	86	129	191	–	0	− 8	− 22	–	0.497
102	123	125	1	0.6	98	314	878	–	86	129	191	–	0	− 8	− 22	–	0.497
102	123	125	1	0.6	98	442	1 005	–	127	212	286	–	0	− 8	− 17	–	0.571
102	123	125	1	0.6	98	493	1 150	–	143	247	334	–	0	− 8	− 17	–	0.497
102	123	125	1	0.6	98	493	1 150	–	143	247	334	–	0	− 8	− 17	–	0.497
104	136	139	1.5	0.8	275	549	1 188	2 348	94	125	176	246	− 9	− 19	− 36	− 58	1.21
104	136	139	1.5	0.8	299	614	1 357	2 712	107	144	205	288	− 9	− 19	− 36	− 58	1.04
104	136	139	1.5	0.8	421	808	1 832	3 786	227	287	392	525	− 7	− 13	− 25	− 42	1.21
104	136	139	1.5	0.8	469	919	2 119	4 417	263	336	460	619	− 7	− 13	− 25	− 42	1.04
104	136	139	1.5	0.8	98	811	2 592	4 157	182	372	569	682	0	− 10	− 25	− 35	1.23
104	136	139	1.5	0.8	98	345	854	–	77	120	170	–	0	− 10	− 24	–	1.30
104	136	139	1.5	0.8	98	380	971	–	86	138	198	–	0	− 10	− 24	–	1.21
104	136	139	1.5	0.8	98	380	971	–	86	138	198	–	0	− 10	− 24	–	1.21
104	136	139	1.5	0.8	98	671	1 381	–	127	246	320	–	0	− 12	− 22	–	1.30
104	136	139	1.5	0.8	98	760	1 590	–	143	287	375	–	0	− 12	− 22	–	1.21
104	136	139	1.5	0.8	98	760	1 590	–	143	287	375	–	0	− 12	− 22	–	1.21
107	158	163	2	1	448	876	2 081	4 153	98	130	192	270	− 18	− 33	− 63	− 99	2.64
107	158	163	2	1	498	995	2 404	4 834	114	151	225	317	− 18	− 33	− 63	− 99	2.18
107	158	163	2	1	703	1 390	3 124	6 301	240	308	419	557	− 13	− 23	− 42	− 68	2.63
107	158	163	2	1	796	1 601	3 639	7 386	280	361	494	657	− 13	− 23	− 42	− 68	2.17
107	158	163	2	1	356	1 633	4 191	6 644	248	422	596	711	− 5	− 20	− 40	− 55	2.67

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

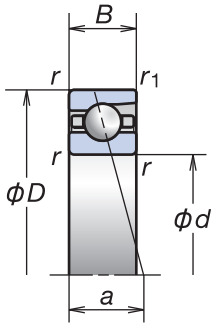
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

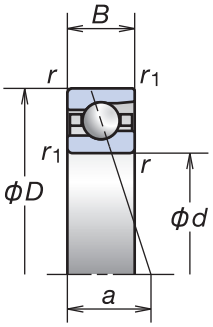
- 동등가 하중 …… P191
- 정등가 하중 …… P198
- 스페이서 · 노즐 목표 위치 …… P237
- 그리스 추천 봉입량 …… P257

1. 앵귤러 볼 베어링

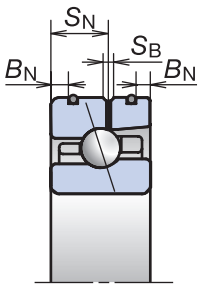
내경100mm



79,70,72계열
BNR10,20계열
BER10,20계열



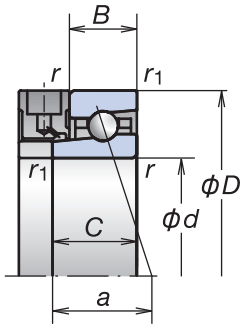
BNR19,29계열
BER19,29계열



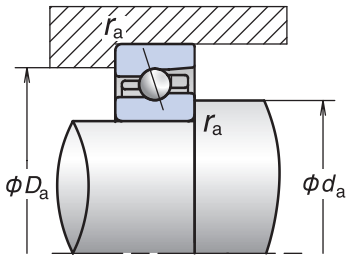
로바스트 쇼트

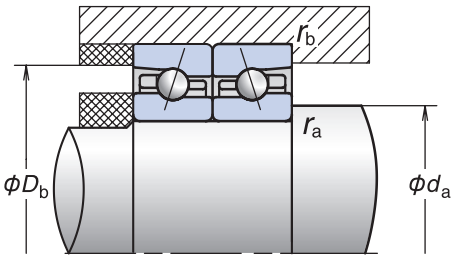
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7920C	100	140	20	–	–	–	–	1.1	0.6	52.5	54.0	33.0	15	16.5	26.1	9 600	14 600
*7920CSN24	100	140	20	–	–	–	–	1.1	0.6	(52.5)	(54.0)	39.2	15	16.5	26.1	12 500	19 000
*7920A5	100	140	20	–	–	–	–	1.1	0.6	49.5	51.5	39.5	25	–	38.0	8 400	12 500
*7920A5SN24	100	140	20	–	–	–	–	1.1	0.6	(49.5)	(51.5)	46.8	25	–	38.0	10 900	16 300
*100BNR19E	100	140	20	–	–	–	–	1.1	0.6	38.0	35.0	50.5	18	10.8	29.5	12 500	17 900
*100BNR19H	100	140	20	–	4.0	12.0	2.2	1.1	0.6	(38.0)	(35.0)	33.0	18	10.8	29.5	15 000	23 400
*100BNR19X	100	140	20	25	4.0	12.0	2.2	1.1	0.6	(38.0)	(35.0)	33.0	18	10.8	29.5	17 500	27 500
*100BER19E	100	140	20	–	–	–	–	1.1	0.6	36.0	33.5	59.5	25	–	38.0	10 800	15 200
*100BER19H	100	140	20	–	4.0	12.0	2.2	1.1	0.6	(36.0)	(33.5)	40.0	25	–	38.0	13 400	20 900
*100BER19X	100	140	20	25	4.0	12.0	2.2	1.1	0.6	(36.0)	(33.5)	40.0	25	–	38.0	15 900	25 000
*100BNR29EV1V	100	140	24	–	–	–	–	1.1	0.6	38.0	35.0	50.5	18	10.8	31.5	12 500	–
*100BNR29HV1V	100	140	24	–	–	–	–	1.1	0.6	(38.0)	(35.0)	33.0	18	10.8	31.5	15 000	–
*100BNR29XV1V	100	140	24	–	–	–	–	1.1	0.6	(38.0)	(35.0)	33.0	18	10.8	31.5	17 500	–
*100BER29EV1V	100	140	24	–	–	–	–	1.1	0.6	36.0	33.5	59.5	25	–	40.0	10 800	–
*100BER29HV1V	100	140	24	–	–	–	–	1.1	0.6	(36.0)	(33.5)	40.0	25	–	40.0	13 400	–
*100BER29XV1V	100	140	24	–	–	–	–	1.1	0.6	(36.0)	(33.5)	40.0	25	–	40.0	15 900	–
*7020C	100	150	24	–	–	–	–	1.5	1	79.0	77.0	49.0	15	16.0	28.7	9 200	14 000
*7020CSN24	100	150	24	–	–	–	–	1.5	1	(79.0)	(77.0)	58.4	15	16.0	28.7	12 000	18 300
*7020A5	100	150	24	–	–	–	–	1.5	1	75.0	73.5	57.5	25	–	41.1	8 000	12 000
*7020A5SN24	100	150	24	–	–	–	–	1.5	1	(75.0)	(73.5)	68.3	25	–	41.1	10 400	15 600
*7020A	100	150	24	–	–	–	–	1.5	1	72.0	70.5	44.5	30	–	48.1	6 000	8 000
*100BNR10E	100	150	24	–	–	–	–	1.5	1	36.0	36.0	52.0	18	10.9	32.3	12 000	17 200
*100BNR10H	100	150	24	–	5.5	14.5	2.2	1.5	1	(36.0)	(36.0)	34.0	18	10.9	32.3	14 400	22 400
*100BNR10X	100	150	24	29	5.5	14.5	2.2	1.5	1	(36.0)	(36.0)	34.0	18	10.9	32.3	16 800	26 400
*100BER10E	100	150	24	–	–	–	–	1.5	1	34.5	34.5	61.0	25	–	41.2	10 400	14 600
*100BER10H	100	150	24	–	5.5	14.5	2.2	1.5	1	(34.5)	(34.5)	41.0	25	–	41.2	12 800	20 000
*100BER10X	100	150	24	29	5.5	14.5	2.2	1.5	1	(34.5)	(34.5)	41.0	25	–	41.2	15 200	24 000
*100BNR20EV1V	100	150	30	–	–	–	–	1.5	1	36.0	36.0	52.0	18	10.9	35.3	12 000	–
*100BNR20HV1V	100	150	30	–	–	–	–	1.5	1	(36.0)	(36.0)	34.0	18	10.9	35.3	14 400	–
*100BNR20XV1V	100	150	30	–	–	–	–	1.5	1	(36.0)	(36.0)	34.0	18	10.9	35.3	16 800	–
*100BER20EV1V	100	150	30	–	–	–	–	1.5	1	34.5	34.5	61.0	25	–	44.2	10 400	–
*100BER20HV1V	100	150	30	–	–	–	–	1.5	1	(34.5)	(34.5)	41.0	25	–	44.2	12 800	–
*100BER20XV1V	100	150	30	–	–	–	–	1.5	1	(34.5)	(34.5)	41.0	25	–	44.2	15 200	–
7220C	100	180	34	–	–	–	–	2.1	1.1	157	127	88.5	15	14.5	35.7	8 300	12 500
7220A5	100	180	34	–	–	–	–	2.1	1.1	149	121	103	25	–	49.6	7 200	10 800
7220A	100	180	34	–	–	–	–	2.1	1.1	144	117	79.5	30	–	57.4	5 400	7 200

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
107	133	135	1	0.6	191	387	905	1 790	84	112	164	230	− 5	− 13	− 28	− 46	0.800
107	133	135	1	0.6	203	427	1 026	2 057	95	129	191	268	− 5	− 13	− 28	− 46	0.702
107	133	135	1	0.6	318	615	1 229	2 546	209	266	346	463	− 5	− 10	− 18	− 31	0.808
107	133	135	1	0.6	348	694	1 410	2 956	241	310	405	545	− 5	− 10	− 18	− 31	0.710
107	133	135	1	0.6	98	329	879	–	72	111	162	–	0	− 10	− 26	–	0.770
107	133	135	1	0.6	98	362	1 000	–	81	128	189	–	0	− 10	− 26	–	0.673
107	133	135	1	0.6	98	362	1 000	–	81	128	189	–	0	− 10	− 26	–	0.673
107	133	135	1	0.6	98	522	1 142	–	120	213	282	–	0	− 10	− 20	–	0.770
107	133	135	1	0.6	98	585	1 311	–	135	247	330	–	0	− 10	− 20	–	0.673
107	133	135	1	0.6	98	585	1 311	–	135	247	330	–	0	− 10	− 20	–	0.673
107	133	135	1	0.6	98	329	879	–	72	111	162	–	0	− 10	− 26	–	0.902
107	133	135	1	0.6	98	362	1 000	–	81	128	189	–	0	− 10	− 26	–	0.805
107	133	135	1	0.6	98	362	1 000	–	81	128	189	–	0	− 10	− 26	–	0.805
107	133	135	1	0.6	98	522	1 142	–	120	213	282	–	0	− 10	− 20	–	0.902
107	133	135	1	0.6	98	585	1 311	–	135	247	330	–	0	− 10	− 20	–	0.805
107	133	135	1	0.6	98	585	1 311	–	135	247	330	–	0	− 10	− 20	–	0.805
109	141	144	1.5	0.8	282	534	1 278	2 572	97	126	187	264	− 9	− 18	− 37	− 60	1.27
109	141	144	1.5	0.8	306	596	1 462	2 974	111	146	218	309	− 9	− 18	− 37	− 60	1.09
109	141	144	1.5	0.8	434	837	2 009	3 948	236	300	418	549	− 7	− 13	− 26	− 42	1.45
109	141	144	1.5	0.8	483	953	2 327	4 609	274	350	491	647	− 7	− 13	− 26	− 42	1.27
109	141	144	1.5	0.8	98	840	2 701	4 338	188	388	595	713	0	− 10	− 25	− 35	1.28
109	141	144	1.5	0.8	98	352	877	–	78	123	175	–	0	− 10	− 24	–	1.34
109	141	144	1.5	0.8	98	389	999	–	88	142	205	–	0	− 10	− 24	–	1.25
109	141	144	1.5	0.8	98	389	999	–	88	142	205	–	0	− 10	− 24	–	1.25
109	141	144	1.5	0.8	98	689	1 423	–	130	254	331	–	0	− 12	− 22	–	1.34
109	141	144	1.5	0.8	98	780	1 639	–	146	296	388	–	0	− 12	− 22	–	1.25
109	141	144	1.5	0.8	98	780	1 639	–	146	296	388	–	0	− 12	− 22	–	1.25
109	141	144	1.5	0.8	98	352	877	–	78	123	175	–	0	− 10	− 24	–	1.65
109	141	144	1.5	0.8	98	389	999	–	88	142	205	–	0	− 10	− 24	–	1.56
109	141	144	1.5	0.8	98	389	999	–	88	142	205	–	0	− 10	− 24	–	1.56
109	141	144	1.5	0.8	98	689	1 423	–	130	254	331	–	0	− 12	− 22	–	1.65
109	141	144	1.5	0.8	98	780	1 639	–	146	296	388	–	0	− 12	− 22	–	1.56
109	141	144	1.5	0.8	98	780	1 639	–	146	296	388	–	0	− 12	− 22	–	1.56
112	168	173	2	1	503	984	2 337	4 700	104	137	202	284	− 20	− 36	− 68	−107	3.18
112	168	173	2	1	776	1 574	3 500	7 110	252	327	442	588	− 14	− 25	− 45	− 73	3.16
112	168	173	2	1	361	1 664	5 052	7 687	253	430	646	758	− 5	− 20	− 45	− 60	3.21

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

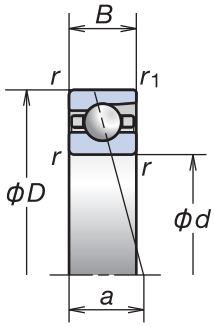
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

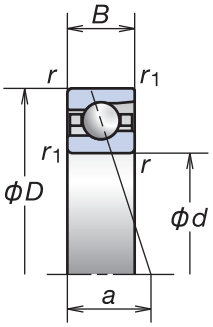
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

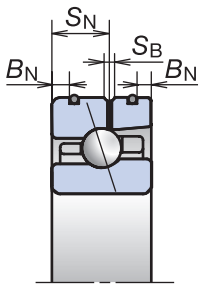
내경105mm



79,70,72계열
BNR10계열
BER10계열



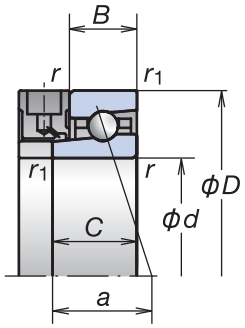
BNR19계열
BER19계열



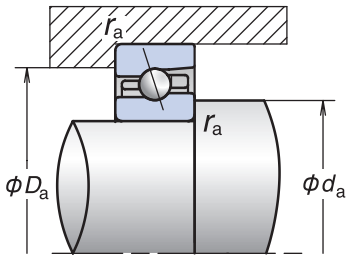
로바스트 쇼트

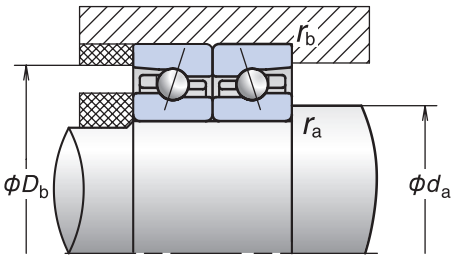
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
7921C	**105**	**145**	**20**	–	–	–	–	1.1	0.6	53.5	57.0	34.5	15	16.6	26.7	9 200	14 000
7921CSN24	**105**	**145**	**20**	–	–	–	–	1.1	0.6	(53.5)	(57.0)	40.8	15	16.6	26.7	12 000	18 300
7921A5	**105**	**145**	**20**	–	–	–	–	1.1	0.6	50.5	54.0	41.0	25	–	39.2	8 000	12 000
7921A5SN24	**105**	**145**	**20**	–	–	–	–	1.1	0.6	(50.5)	(54.0)	48.7	25	–	39.2	10 400	15 600
105BNR19E	**105**	**145**	**20**	–	–	–	–	1.1	0.6	38.5	36.5	53.0	18	10.8	30.3	12 000	17 200
105BNR19H	**105**	**145**	**20**	–	–	–	–	1.1	0.6	(38.5)	(36.5)	39.0	18	10.8	30.3	14 400	22 400
105BNR19X	**105**	**145**	**20**	25	–	–	–	1.1	0.6	(38.5)	(36.5)	39.0	18	10.8	30.3	16 800	26 400
105BER19E	**105**	**145**	**20**	–	–	–	–	1.1	0.6	37.0	35.0	62.0	25	–	39.2	10 400	14 600
105BER19H	**105**	**145**	**20**	–	–	–	–	1.1	0.6	(37.0)	(35.0)	42.0	25	–	39.2	12 800	20 000
105BER19X	**105**	**145**	**20**	25	–	–	–	1.1	0.6	(37.0)	(35.0)	42.0	25	–	39.2	15 200	24 000
7021C	**105**	**160**	**26**	–	–	–	–	2	1	92.5	89.5	57.0	15	15.9	30.7	8 700	13 300
7021CSN24	**105**	**160**	**26**	–	–	–	–	2	1	(92.5)	(89.5)	68.1	15	15.9	30.7	11 400	17 300
7021A5	**105**	**160**	**26**	–	–	–	–	2	1	87.5	85.0	66.5	25	–	43.9	7 600	11 400
7021A5SN24	**105**	**160**	**26**	–	–	–	–	2	1	(87.5)	(85.0)	79.0	25	–	43.9	9 900	14 800
7021A	**105**	**160**	**26**	–	–	–	–	2	1	84.0	81.5	51.0	30	–	51.2	5 700	7 600
105BNR10E	**105**	**160**	**26**	–	–	–	–	2	1	41.0	41.0	59.5	18	10.9	34.5	11 400	16 200
105BNR10H	**105**	**160**	**26**	–	6.0	15.2	2.2	2	1	(41.0)	(41.0)	39.0	18	10.9	34.5	13 600	21 200
105BNR10X	**105**	**160**	**26**	31	6.0	15.2	2.2	2	1	(41.0)	(41.0)	39.0	18	10.9	34.5	15 900	25 000
105BER10E	**105**	**160**	**26**	–	–	–	–	2	1	39.0	39.5	70.0	25	–	43.9	9 800	13 800
105BER10H	**105**	**160**	**26**	–	6.0	15.2	2.2	2	1	(39.0)	(39.5)	47.5	25	–	43.9	12 100	18 900
105BER10X	**105**	**160**	**26**	31	6.0	15.2	2.2	2	1	(39.0)	(39.5)	47.5	25	–	43.9	14 400	22 700
7221C	**105**	**190**	**36**	–	–	–	–	2.1	1.1	171	143	97.5	15	14.5	37.7	7 800	11 900
7221A5	**105**	**190**	**36**	–	–	–	–	2.1	1.1	163	137	111	25	–	52.4	6 800	10 200
7221A	**105**	**190**	**36**	–	–	–	–	2.1	1.1	157	132	85.0	30	–	60.6	5 100	6 800

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
112	138	140	1	0.6	194	396	890	1 791	86	116	167	235	− 5	− 13	− 27	− 45	0.831
112	138	140	1	0.6	206	437	1 009	2 058	98	133	193	274	− 5	− 13	− 27	− 45	0.729
112	138	140	1	0.6	272	565	1 361	2 760	203	265	369	491	− 4	− 9	− 19	− 32	0.820
112	138	140	1	0.6	296	636	1 565	3 208	234	308	432	577	− 4	− 9	− 19	− 32	0.718
112	138	140	1	0.6	98	336	906	–	74	115	168	–	0	− 10	− 26	–	0.795
112	138	140	1	0.6	98	370	1 032	–	83	133	196	–	0	− 10	− 26	–	0.693
112	138	140	1	0.6	98	370	1 032	–	83	133	196	–	0	− 10	− 26	–	0.693
112	138	140	1	0.6	98	536	1 180	–	124	220	293	–	0	− 10	− 20	–	0.795
112	138	140	1	0.6	98	602	1 355	–	139	256	343	–	0	− 10	− 20	–	0.693
112	138	140	1	0.6	98	602	1 355	–	139	256	343	–	0	− 10	− 20	–	0.693
115	150	154	2	1	330	625	1 494	2 973	103	134	197	276	− 11	− 21	− 42	− 67	1.58
115	150	154	2	1	362	703	1 715	3 446	118	155	230	324	− 11	− 21	− 42	− 67	1.34
115	150	154	2	1	493	989	2 211	4 473	247	318	431	572	− 8	− 15	− 28	− 46	1.82
115	150	154	2	1	553	1 131	2 564	5 228	287	372	507	674	− 8	− 15	− 28	− 46	1.58
115	150	154	2	1	98	840	2 693	4 318	188	388	591	707	0	− 10	− 25	− 35	1.60
115	150	154	2	1	98	424	1 034	–	80	135	190	–	0	− 12	− 27	–	1.70
115	150	154	2	1	98	471	1 182	–	89	156	222	–	0	− 12	− 27	–	1.59
115	150	154	2	1	98	471	1 182	–	89	156	222	–	0	− 12	− 27	–	1.59
115	150	154	2	1	98	911	1 815	–	133	286	369	–	0	− 15	− 26	–	1.70
115	150	154	2	1	98	1 039	2 100	–	149	335	434	–	0	− 15	− 26	–	1.59
115	150	154	2	1	98	1 039	2 100	–	149	335	434	–	0	− 15	− 26	–	1.59
117	178	183	2	1	540	1 077	2 651	5 311	108	144	216	304	− 21	− 38	− 73	−114	3.78
117	178	183	2	1	862	1 790	3 950	7 929	268	348	471	625	− 15	− 27	− 48	− 77	3.77
117	178	183	2	1	368	2 298	5 208	8 924	260	493	666	818	− 5	− 25	− 45	− 65	3.82

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

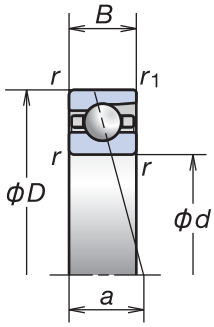
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

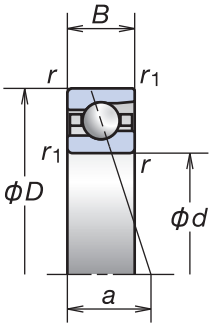
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

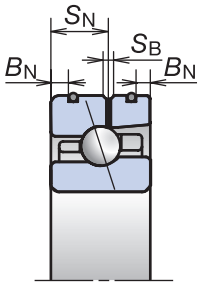
내경110mm



79,70,72계열
BNR10계열
BER10계열



BNR19계열
BER19계열



로바스트 쇼트

호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
7922C	110	150	20	–	–	–	–	1.1	0.6	54.5	59.5	35.5	15	16.7	27.4	8 900	13 500
7922CSN24	110	150	20	–	–	–	–	1.1	0.6	(54.5)	(59.5)	42.4	15	16.7	27.4	11 600	17 600
7922A5	110	150	20	–	–	–	–	1.1	0.6	51.5	56.0	43.0	25	–	40.3	7 700	11 600
7922A5SN24	110	150	20	–	–	–	–	1.1	0.6	(51.5)	(56.0)	50.7	25	–	40.3	10 000	15 000
*110BNR19E	110	150	20	–	–	–	–	1.1	0.6	39.0	38.0	55.5	18	10.9	31.1	11 600	16 500
*110BNR19H	110	150	20	–	4.0	12.0	2.2	1.1	0.6	(39.0)	(38.0)	42.0	18	10.9	31.1	13 900	21 600
*110BNR19X	110	150	20	25	4.0	12.0	2.2	1.1	0.6	(39.0)	(38.0)	42.0	18	10.9	31.1	16 200	25 400
*110BER19E	110	150	20	–	–	–	–	1.1	0.6	37.5	36.5	65.0	25	–	40.3	10 000	14 000
*110BER19H	110	150	20	–	4.0	12.0	2.2	1.1	0.6	(37.5)	(36.5)	44.0	25	–	40.3	12 400	19 300
*110BER19X	110	150	20	25	4.0	12.0	2.2	1.1	0.6	(37.5)	(36.5)	44.0	25	–	40.3	14 700	23 100
7022C	110	170	28	–	–	–	–	2	1	111	104	68.5	15	15.6	32.7	8 300	12 500
7022CSN24	110	170	28	–	–	–	–	2	1	(111)	(104)	81.0	15	15.6	32.7	10 800	16 300
7022A5	110	170	28	–	–	–	–	2	1	105	99.0	79.5	25	–	46.6	7 200	10 800
7022A5SN24	110	170	28	–	–	–	–	2	1	(105)	(99.0)	94.1	25	–	46.6	9 300	14 000
7022A	110	170	28	–	–	–	–	2	1	101	95.5	61.0	30	–	54.4	5 400	7 200
110BNR10E	110	170	28	–	–	–	–	2	1	46.0	47.0	68.0	18	10.9	36.7	10 800	15 300
110BNR10H	110	170	28	–	6.0	16.2	2.2	2	1	(46.0)	(47.0)	44.5	18	10.9	36.7	12 900	20 000
110BNR10X	110	170	28	33	6.0	16.2	2.2	2	1	(46.0)	(47.0)	44.5	18	10.9	36.7	15 000	23 600
110BER10E	110	170	28	–	–	–	–	2	1	44.0	45.0	79.5	25	–	46.7	9 300	13 000
110BER10H	110	170	28	–	6.0	16.2	2.2	2	1	(44.0)	(45.0)	54.0	25	–	46.7	11 500	17 900
110BER10X	110	170	28	33	6.0	16.2	2.2	2	1	(44.0)	(45.0)	54.0	25	–	46.7	13 600	21 500
7222C	110	200	38	–	–	–	–	2.1	1.1	185	160	108	15	14.5	39.8	7 500	11 300
7222A5	110	200	38	–	–	–	–	2.1	1.1	176	153	126	25	–	55.1	6 500	9 700
7222A	110	200	38	–	–	–	–	2.1	1.1	170	148	97.0	30	–	63.7	4 900	6 500

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.